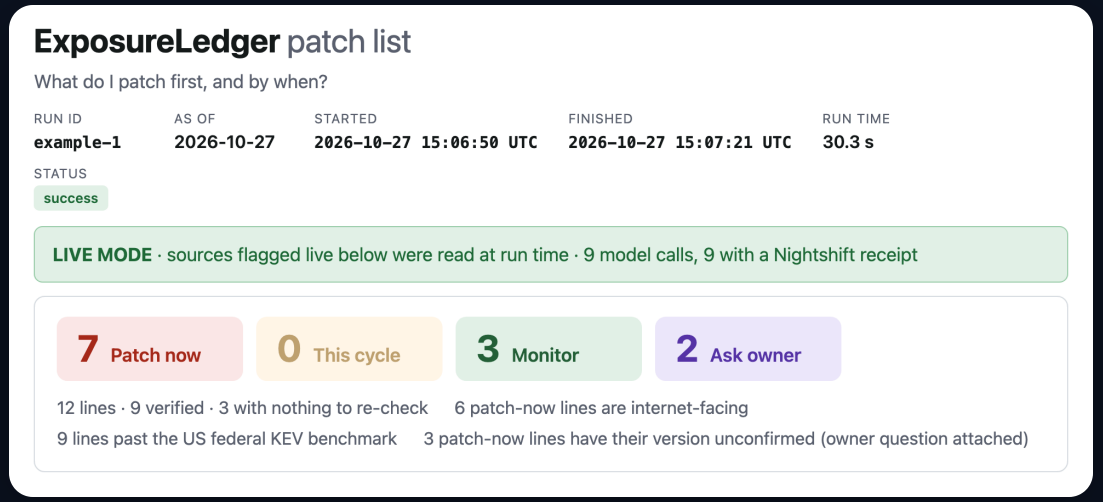
OPEN AGENT HACKATHON 2026

# ExposureLedger

## What do I patch first, and by when?

Agents that turn a small organisation's messy software list into a dated, evidence-backed patch list.

**Track: Solving Fragmented Intelligence**




**A real live run** (example 1, 27 Oct 2026): 12 rows of the synthetic software list in 30.3 s.

---

**Team Megatron's Interns:** Mohd Amir Suhel, Thant Zaw Latt, Berkin Kaynar

**Code:** github.com/berkinksk/Open-Agent-Hackathon-2026

FINAL TEST **0 wrong answers** on 30 held-out rows (one run):
23 of 23 answered rows right; 7 sent to the owner with a question instead of a decision.

Built on Zetaris · Meterless World Model engine · Meterless Nightshift · NVIDIA Nemotron

POINT 1 · THE PROBLEM AND THE USERS

# One IT person, four sources, one deadline

**The only IT admin** at a ~200-person charity.
No security team.

**"What do I patch first, and by when?"**

**Their software list**
free text, typed by hand

```
MS Exchange Svr
2019 CU12
```

name match? →

**CISA KEV**
exploited in the wild

```
Microsoft /
Exchange Server
```

CVE ID →

**FIRST EPSS**
CVE ID only

```
CVE-2022-41082
→ 0.9997
```

CVE ID →

**NIST NVD**
CVE + affected versions

```
exchange_server:2019:
cumulative_update_12
```

One mail server, four names. Synthetic software list; real KEV, EPSS and NVD data (example 1).

POINT 2 · THE SOLUTION

# From a messy list to a dated patch list

**IN: THE LIST AS WRITTEN**

- `MS Exchange Svr 2019 CU12`
- `Apache HTTPD 2.4.49 / public website`
- `Citrix netscaler ADC / Gateway; firmware tbc`
- `Windows Server 2019 std (mail host)`
- `Microsoft Word Online (browser app)`
- `7-Zip v24.09 (64 bit)`

+ 6 more rows (example 1)

→

**ExposureLedger**

30.3 s · 9 model calls

## Patch now 7 lines

`public-web-01` · Apache 2.4.49 → **patch by 2026-11-03**

## Ask the owner 2 lines

`mail-01` · "Which exact Windows version or build runs on mail-01?"

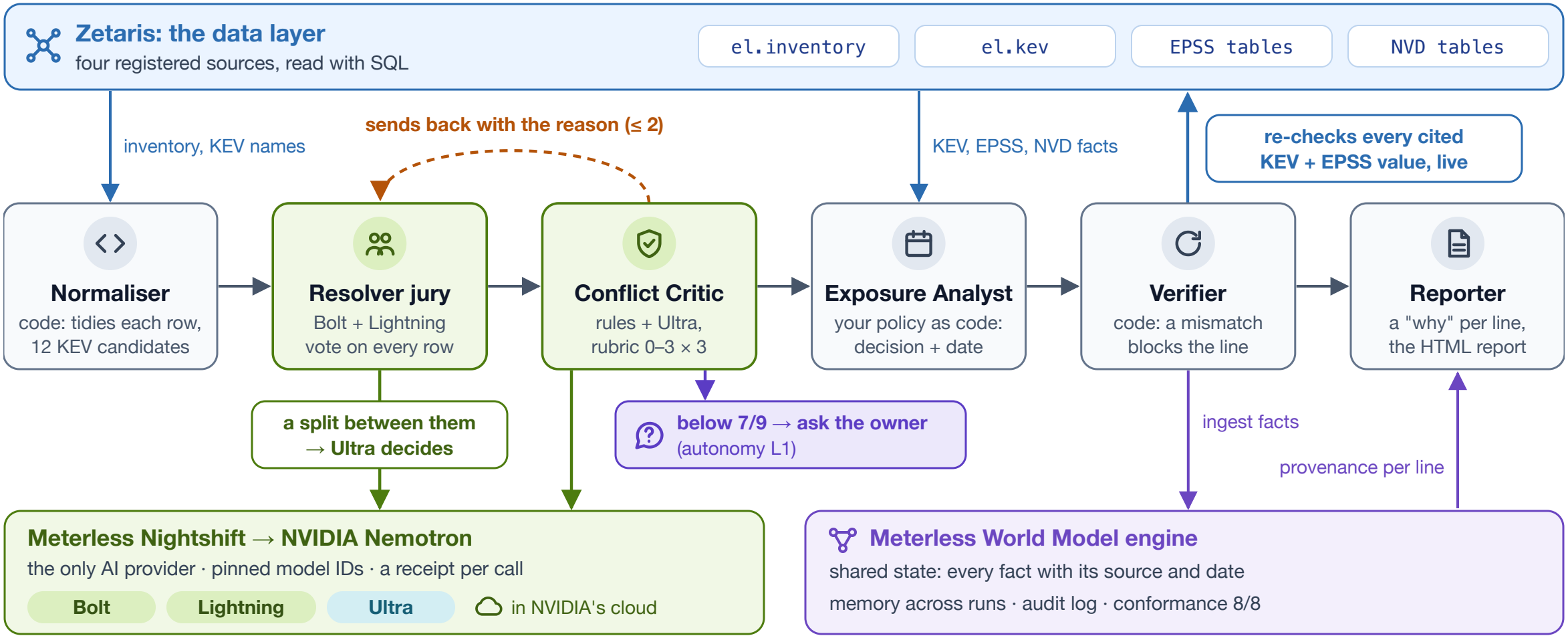
## Monitor 3 lines

`fin-laptop-01` · 7-Zip 24.09 → **review by 2027-01-25**

**Decision support.** Dates follow the user's own rules (patch now = 7 days by default). No claim that a flaw is exploitable on their systems. The software list is synthetic.

POINT 3 · THE AGENT WORKFLOW

# Six agents that check each other



POINT 4 · THE TECHNICAL STACK

# Small, zero dependencies, runs from a clean clone

**4 sources** — software list (synthetic, GitHub raw) · CISA KEV · FIRST EPSS · NIST NVD

read in place

**Zetaris** — local Docker · REST SQL API · one query at a time

named SQL queries

**Six agents** — TypeScript on Node.js 25 · no npm packages

Normaliser · Jury · Critic · Analyst · Verifier · Reporter

**World Model**

Meterless engine, vendored unmodified

model calls

**Meterless Nightshift** — local · OpenAI-compatible · a receipt per call

pinned model IDs

**NVIDIA Nemotron** — Bolt · Lightning · Ultra, in NVIDIA's cloud, reached through Nightshift

### RUN IT

`npm run cli` · `POST /run`

a Docker image

**Sample mode:** offline, no keys

### CHECK IT

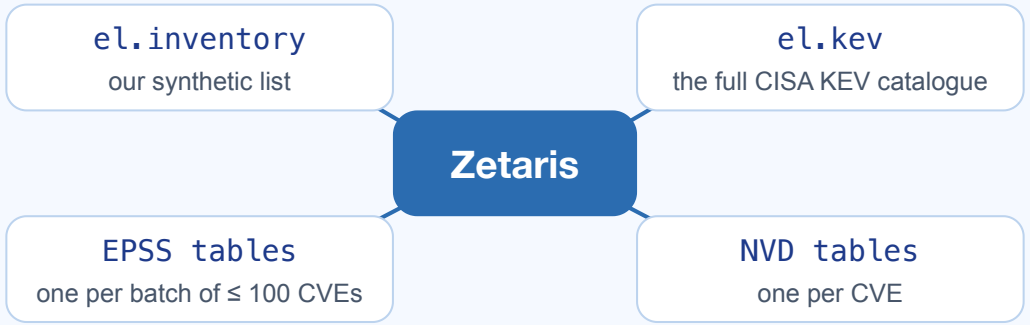
- ✓ `npm test`
- ✓ `npm run verify`
- ✓ `verify:engine` 4/4
- ✓ CI on every push

Built with Claude Code; Cursor not used.

# Zetaris is the data layer; Meterless is the engine and the AI provider

## Zetaris: one hub, four sources

Named queries only; EPSS and NVD tables are made during the run.



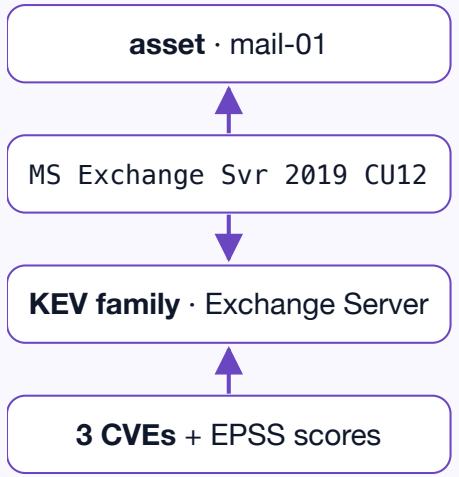

```
SELECT … FROM el.inventory i JOIN el.kev k
ON instr(lower(i.software), lower(trim(k.product))) > 0
```

**One federated SQL, display only: hits 9 of 12 rows in 921 ms**

Example 1: 13 queries + 2 table registrations, 5.68 s in Zetaris.

## World Model: every fact has a source

The Exchange line of example 1, as the engine stores it:




**installed_on**
zetaris:el.inventory · 2026-10-27

**resolves_to** · agent:jury, receipts
`rcpt_req_m0ofmgp1, rcpt_req_wxor4tk6`

**affects**
zetaris:el.kev · catalogue 2026.10.26

**EPSS** · zetaris:epss · 2026-10-27

✓ `verify:engine 4/4 · conformance 8/8`

**Nightshift** is the only AI provider: pinned model IDs, a receipt for every call.

POINT 6 · WHAT WAS UNIQUE ABOUT OUR USE OF THE SPONSOR TECHNOLOGIES

# Disagreement as a signal, and a live re-check before anything is shown

## Disagreement calls in Ultra

Example 1: 3 splits; Ultra decided all 3, each with a receipt.

## A critic blocks look-alikes

"Word Online" is not installed Word: sent back twice, then the owner is asked.

## Live re-check in Zetaris

Every cited KEV entry and EPSS score is read again before a line is shown.

## Fail-closed by design

KEV or EPSS unreadable → the line goes to the owner, not to "monitor".

POINT 7 · PRODUCT VISUALS

# What the admin sees: a date, a deadline and a reason

| DECISION · ASSET | SOFTWARE (AS WRITTEN) | RESOLVED KEV PRODUCT | KEV DUE (US FEDERAL BENCHMARK) 1 | TARGET (YOUR RULE) 2 | MAX EPSS | VERIFIED 3 | WHY 4 |
|---|---|---|---|---|---|---|---|
| Patch now **7 lines · house target: within 7 days** | | | | | | | |
| Patch now<br>**public-web-01**<br>A08<br>internet-facing | Apache HTTPD 2.4.49 / public website<br>row R61 | **Apache / HTTP Server**<br>family: apache:http-server<br>version: 2.4.49 · type: server | **2021-11-17**<br>**overdue by 1805 days**<br>3 KEV CVEs | **2026-11-03**<br>patch by (house target) | **0.99992**<br>CVE-2021-41773<br>percentile 99.9 | verified<br>1 note | NVD places Apache HTTP Server 2.4.49 inside the affected ranges of 3 KEV-listed CVEs, most recently CVE-2024-38475, which CISA KEV lists as exploited in the wild (added 2025-05-01); it is outside the ranges of 1 other KEV CVE. Patch by 2026-11-03 under your 7-day rule; the US federal due date (2021-11-17) passed 1805 days ago, and EPSS for CVE-2021-41773 is 0.99992. |

1 **US federal due date**
1,805 days overdue (benchmark)

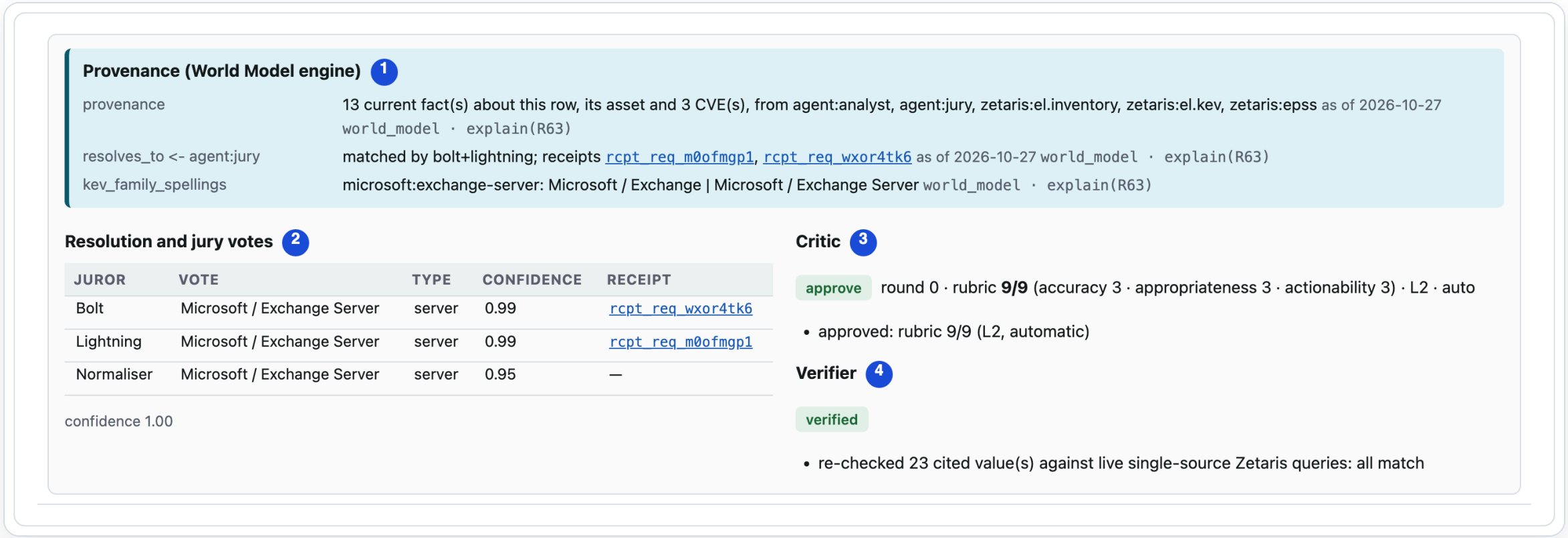
2 **Your rule**
patch by 2026-11-03

3 **Re-checked live**
in Zetaris before it is shown

4 **A plain "why"**
written from the facts

POINT 7 · PRODUCT VISUALS

# Open any line: where each fact came from, and who decided



**Provenance (World Model engine)** 1

| | |
|---|---|
| provenance | 13 current fact(s) about this row, its asset and 3 CVE(s), from agent:analyst, agent:jury, zetaris:el.inventory, zetaris:el.kev, zetaris:epss as of 2026-10-27 `world_model · explain(R63)` |
| resolves_to <- agent:jury | matched by bolt+lightning; receipts rcpt_req_m0ofmgp1, rcpt_req_wxor4tk6 as of 2026-10-27 `world_model · explain(R63)` |
| kev_family_spellings | microsoft:exchange-server: Microsoft / Exchange \| Microsoft / Exchange Server `world_model · explain(R63)` |

**Resolution and jury votes** 2

| JUROR | VOTE | TYPE | CONFIDENCE | RECEIPT |
|---|---|---|---|---|
| Bolt | Microsoft / Exchange Server | server | 0.99 | rcpt_req_wxor4tk6 |
| Lightning | Microsoft / Exchange Server | server | 0.99 | rcpt_req_m0ofmgp1 |
| Normaliser | Microsoft / Exchange Server | server | 0.95 | — |

confidence 1.00

**Critic** 3

approve round 0 · rubric **9/9** (accuracy 3 · appropriateness 3 · actionability 3) · L2 · auto

- approved: rubric 9/9 (L2, automatic)

**Verifier** 4

verified

- re-checked 23 cited value(s) against live single-source Zetaris queries: all match

1 **Provenance**
from the World Model engine

2 **Jury votes**
each with its Nightshift receipt

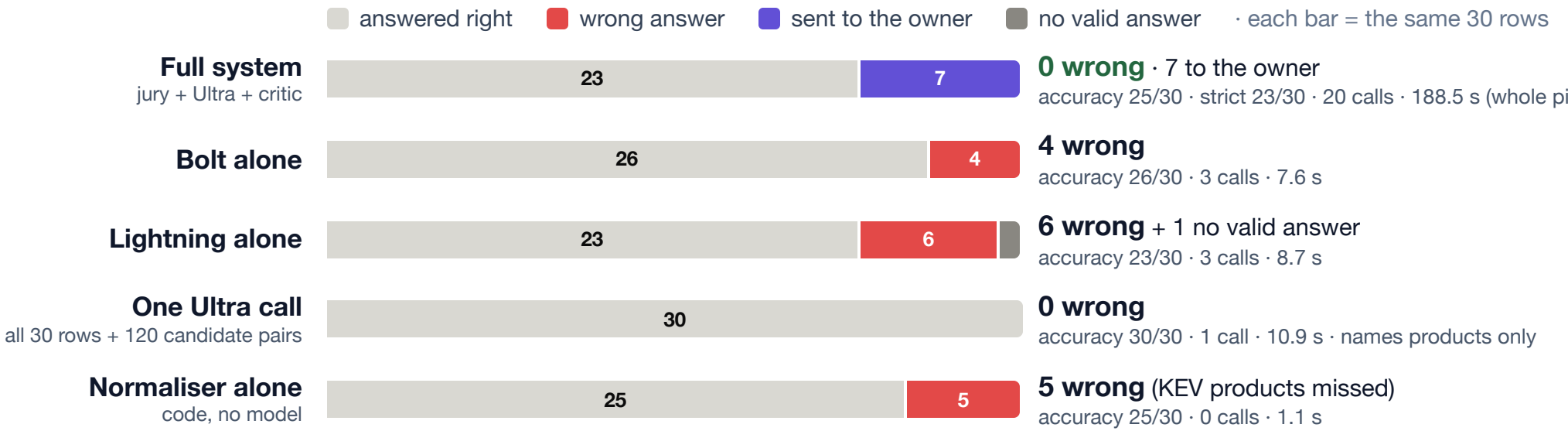
3 **Critic**
rubric 9/9, approved

4 **Verifier**
23 cited values re-checked live

RESULTS

# Measured once, reported as it is

**0 wrong answers on 30 held-out rows (one run): 23 of 23 answered rows right; 7 sent to the owner with a question instead of a decision.**




**The 7:** 4 named the right product and asked only for the version or build; 1 Windows Server row was held back by the critic's operating-system rule; 2 were look-alikes the critic stopped.

Answers written independently by two teammates; where they differed, one of their two answers was chosen per row by the labelling rules (`data/test_answers.csv`). Rows: a seeded random pick of 30 from the 80-row synthetic list, sealed during development; run once on 27 Oct 2026, on commit `e1edc33`; scored by KEV product family. *Accuracy* counts a row left to the owner as right only when the answer is "no KEV product"; *strict* never does. None of the 18 rows with a KEV product got "no KEV product" as its answer; none of the 12 without one got a KEV product as its answer. 3 rows (R13, R46, R68) resemble examples the rules were built on; on the other 27: 0 wrong, 20 of 20 answered rows right, 7 to the owner. After the test, the normaliser was changed to read glued Windows spellings ("Windows11", "Win Server"); the test was not re-run. Source: `docs/eval/final-test/eval-2026-10-27-152613-test.json`, `docs/evaluation.md`.

RESULTS · THE ONE-CALL BASELINE, SHOWN HONESTLY

# One Ultra call matched 30 of 30; naming is one step of five

**30/30**

**on this test**

**1 Ultra call** · 10.9 s
all 30 rows + the normaliser's 120 candidate pairs

It names products only.

**Why the full system is still better:** a name is not a decision. One call checks no version, re-reads no evidence, sets no date and asks no question; the full system does all four.

| WHAT A PATCH DECISION NEEDS | ONE ULTRA CALL | FULL SYSTEM |
|---|---|---|
| Name the KEV product, or none | ✓ | ✓ |
| Check the version against NVD's ranges | – | ✓ |
| Re-read every cited KEV and EPSS value live | – | ✓ |
| Set a dated decision under the user's rules | – | ✓ |
| Ask the owner when the evidence is thin | – | ✓ |
| Cite the evidence and receipts per line | – | ✓ |

**No single model is trusted alone:** in our first probe (26 Oct), all three models, Ultra included, matched "Exchange Online" to Exchange Server and "FortiClient VPN" to FortiClient EMS.

NVIDIA NEMOTRON

# Three Nemotron sizes, three jobs, measured

## Bolt

`nvidia/nemotron-3.5-bolt-e4b`

First vote on every row

**3**
calls, final test

## Lightning

`nvidia/nemotron-3.5-lightning-30b-a3b`

Second vote; first re-decide

**5**
calls, final test

## Ultra

`nvidia/nemotron-3-ultra-550b-a55b`

Decides splits, critiques, re-decides

**12**
calls, final test

**8 of 30**
rows needed Ultra

### EXAMPLE 1: EVERY CALL WITH ITS NIGHTSHIFT RECEIPT (REQUESTED MODEL = ROUTED MODEL)

| | | |
|---|---|---|
| `jury.lightning` Lightning<br>`rcpt_req_juzzifxr` | `jury.bolt` Bolt<br>`rcpt_req_oj3cxr4j` | `jury.lightning` Lightning<br>`rcpt_req_m0ofmgp1` |
| `jury.bolt` Bolt<br>`rcpt_req_wxor4tk6` | `jury.adjudicate` Ultra<br>`rcpt_req_n06lqy7s` | `critic` Ultra<br>`rcpt_req_xih128zv` |
| `critic` Ultra<br>`rcpt_req_0sosksov` | `resolver.retry.lightning` Lightning<br>`rcpt_req_zw1k6iqd` | `resolver.retry.ultra` Ultra<br>`rcpt_req_xf1qqdk8` |

In the three example runs, **15 of 15 calls** were answered by the pinned model we asked for.

POINT 8 · FUTURE ENHANCEMENTS

# What comes next

NEXT 1

## Switch on the NVD product lookup

Fills "this cycle"; built, but off in this build

NEXT 2

## Ultra-first matching

Bolt and Lightning as quick cross-checks

NEXT 3

## Reopen decisions automatically

When KEV changes or the owner answers

NEXT 4

## More sources and reach

Vendor advisories, tickets, daily runs

TRY IT OFFLINE: NO KEYS, NO NETWORK

```
npm run cli -- --input input_examples/example_2.json --sample
```

Node.js 25 or newer; nothing to install.

**Code:** github.com/berkinksk/Open-Agent-Hackathon-2026

**Team Megatron's Interns:** Mohd Amir Suhel, Thant Zaw Latt, Berkin Kaynar